

Session 7:

Task 1:

Process Synchronization and Child Process Monitoring Using waitpid()



```
GNU nano 8.7.1 src/task1_waitpid.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main(void)
{
    pid_t pid;
    int status;

    printf("=== Process Synchronization using waitpid() ===\n");
    printf("Parent Process PID: %d\n", getpid());

    pid = fork();

    if (pid < 0)
    {
        perror("fork");
        return 1;
    }

    if (pid == 0)
    {
        /* Child process */
        printf("\nChild Process Started\n");
        printf("Child PID: %d\n", getpid());

        printf("Child is working...\n");
        sleep(2);

        printf("Child Process Completed\n");

        exit(10);
    }
    else
    {
        /* Parent process */
        printf("Parent Process Started\n");
        printf("Parent PID: %d\n", getpid());

        printf("Waiting for child process...\n");
        waitpid(pid, &status, 0);

        printf("Child Process Completed\n");
        exit(10);
    }
}
```

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```

/* Parent process */
printf("\nParent is waiting for Child PID: %d\n", pid);

if (waitpid(pid, &status, 0) == -1)
{
    perror("waitpid");
    return 1;
}

printf("\nParent detected child termination.\n");

if (WIFEXITED(status))
{
    printf("Child exited normally.\n");
    printf("Child Exit Status: %d\n",
           WEXITSTATUS(status));
}
else if (WIFSIGNALED(status))
{
    printf("Child terminated by signal: %d\n",
           WTERMSIG(status));
}

printf("Parent Process Completed\n");
}

return 0;
}

```

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```

jella_pragathi@LAPTOP-RQQGT555:~$ cd ~/ossp_project
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ pwd
/home/jella_pragathi/ossp_project
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ mkdir -p ~/ossp_project/src ~
/ossp_project/build
cd ~/ossp_project
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ nano src/task1_waitpid.c
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ gcc -Wall -Wextra -g src/task
1_waitpid.c -o build/task1_waitpid
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ ./build/task1_waitpid
=== Process Synchronization using waitpid() ===
Parent Process PID: 1200

Parent is waiting for Child PID: 1201

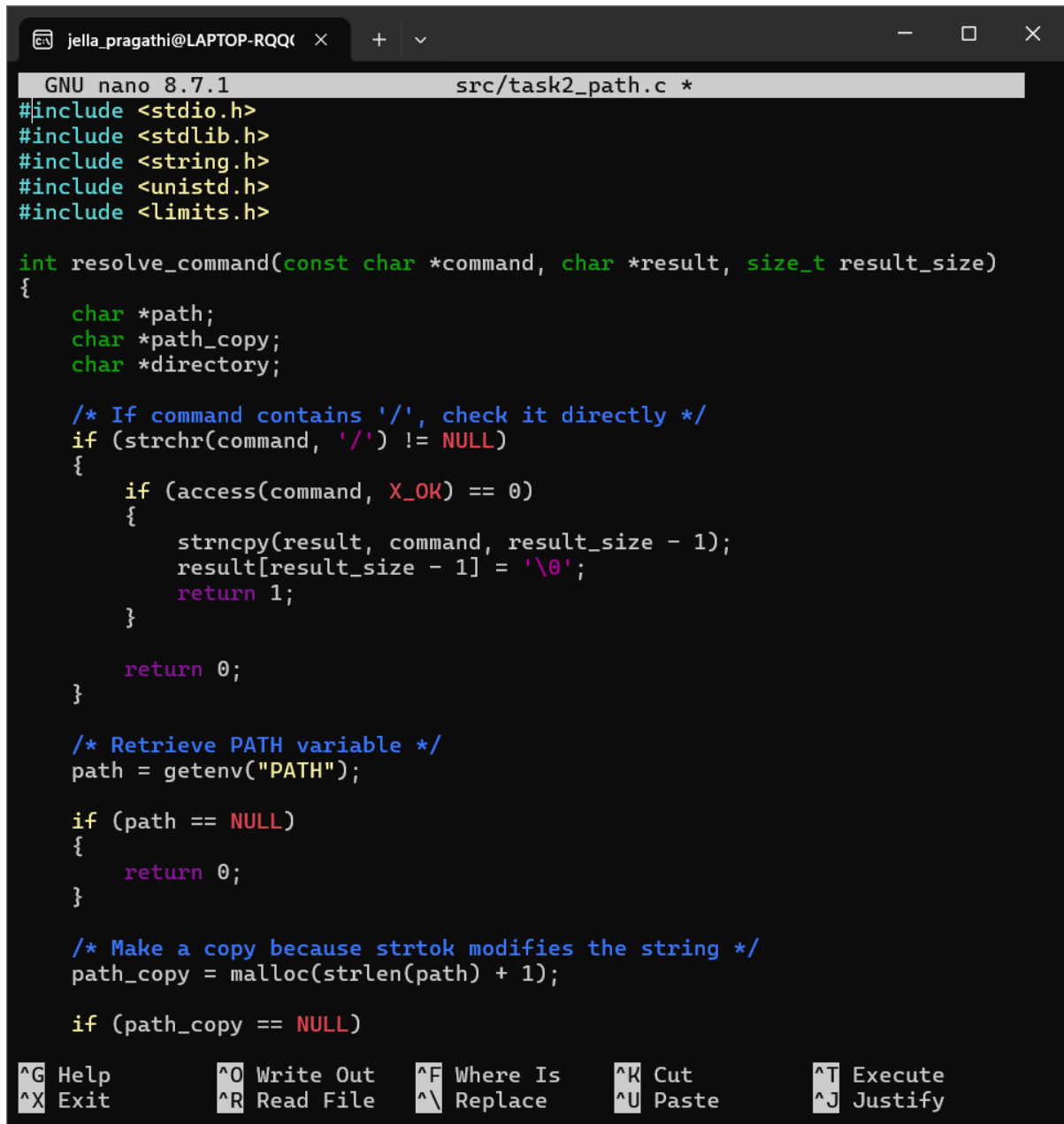
Child Process Started
Child PID: 1201
Child is working...
Child Process Completed

Parent detected child termination.
Child exited normally.
Child Exit Status: 10
Parent Process Completed

```

Task 2:

To Retrieve PATH Variable, Parse Search Directories, Locate Executables, Verify Permissions, Handle Missing Commands, Test Command Resolution



```
GNU nano 8.7.1 src/task2_path.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>

int resolve_command(const char *command, char *result, size_t result_size)
{
    char *path;
    char *path_copy;
    char *directory;

    /* If command contains '/', check it directly */
    if (strchr(command, '/') != NULL)
    {
        if (access(command, X_OK) == 0)
        {
            strncpy(result, command, result_size - 1);
            result[result_size - 1] = '\0';
            return 1;
        }

        return 0;
    }

    /* Retrieve PATH variable */
    path = getenv("PATH");

    if (path == NULL)
    {
        return 0;
    }

    /* Make a copy because strtok modifies the string */
    path_copy = malloc(strlen(path) + 1);

    if (path_copy == NULL)
```

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GNU nano 8.7.1

src/task2_path.c *

```
if (path_copy == NULL)
{
    perror("malloc");
    return 0;
}

strcpy(path_copy, path);

/* Split PATH into directories */
directory = strtok(path_copy, ":");

while (directory != NULL)
{
    char candidate[PATH_MAX];

    /* Build complete executable path */
    snprintf(candidate,
              sizeof(candidate),
              "%s/%s",
              directory,
              command);

    /* Check whether file exists and is executable */
    if (access(candidate, X_OK) == 0)
    {
        strncpy(result, candidate, result_size - 1);
        result[result_size - 1] = '\0';

        free(path_copy);
        return 1;
    }

    directory = strtok(NULL, ":");
}

free(path_copy);
```

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```
jella_pragathi@LAPTOP-RQQ... × + ▾
GNU nano 8.7.1 src/task2_path.c *
    return 0;
}

int main(int argc, char *argv[])
{
    char result[PATH_MAX];

    printf("=== PATH Command Resolution ===\n\n");

    /* Check command-line argument */
    if (argc != 2)
    {
        printf("Usage: %s <command>\n", argv[0]);
        return 1;
    }

    printf("Command to resolve: %s\n", argv[1]);

    /* Display PATH */
    printf("\nPATH variable:\n%s\n", getenv("PATH"));

    /* Resolve command */
    printf("\nSearching PATH...\n");

    if (resolve_command(argv[1], result, sizeof(result)))
    {
        printf("Command found!\n");
        printf("Executable path: %s\n", result);
    }
    else
    {
        printf("Command not found: %s\n", argv[1]);
    }

    return 0;
}
|

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^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

```
jella_pragathi@LAPTOP-RQQI:~$ nano src/task2_path.c
jella_pragathi@LAPTOP-RQQI:~$ gcc -Wall -Wextra -g src/task2_path.c -o build/task2_path
jella_pragathi@LAPTOP-RQQI:~$ ./build/task2_path ls
=== PATH Command Resolution ===

Command to resolve: ls

PATH variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporat
ionII.WindowsSubsystemForLinux.2.7.11.0_x64_8wekyb3d8bbwe:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Or
acle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/windows/system32:/mnt/c/windows:/mnt/c/windows/System32/Wbem:/mnt/c/
windows/System32/WindowsPowerShell/v1.0:/mnt/c/windows/System32/OpenSSH:/mnt/c/Users/jella_pragathi/Downloads/gcc-15.1.0/bin:/mnt/c/Program Files/Git/cmd:/
mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/jella_pragathi/AppData/Local/Programs/Python/Python314/Scripts:/m
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hi/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/jella_pragathi/AppData/Roaming/npm:/snap/bin

Searching PATH...
Command found!
Executable path: /usr/bin/ls
jella_pragathi@LAPTOP-RQQI:~$ ./build/task2_path gcc
=== PATH Command Resolution ===

Command to resolve: gcc

PATH variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporat
ionII.WindowsSubsystemForLinux.2.7.11.0_x64_8wekyb3d8bbwe:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Or
acle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/windows/system32:/mnt/c/windows:/mnt/c/windows/System32/Wbem:/mnt/c/
windows/System32/WindowsPowerShell/v1.0:/mnt/c/windows/System32/OpenSSH:/mnt/c/Users/jella_pragathi/Downloads/gcc-15.1.0/bin:/mnt/c/Program Files/Git/cmd:/
mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/jella_pragathi/AppData/Local/Programs/Python/Python314/Scripts:/m
nt/c/Users/jella_pragathi/AppData/Local/Programs/Python/Python314:/mnt/c/Users/jella_pragathi/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/jella_pragat
hi/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/jella_pragathi/AppData/Roaming/npm:/snap/bin

Searching PATH...
Command found!
Executable path: /usr/bin/gcc
jella_pragathi@LAPTOP-RQQI:~$ ./build/task2_path gcc
```

```
jella_pragathi@LAPTOP-RQQI:~$ Executable path: /usr/bin/gcc
jella_pragathi@LAPTOP-RQQI:~$ ./build/task2_path hello123
=== PATH Command Resolution ===

Command to resolve: hello123

PATH variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporat
ionII.WindowsSubsystemForLinux.2.7.11.0_x64_8wekyb3d8bbwe:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Or
acle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/windows/system32:/mnt/c/windows:/mnt/c/windows/System32/Wbem:/mnt/c/
windows/System32/WindowsPowerShell/v1.0:/mnt/c/windows/System32/OpenSSH:/mnt/c/Users/jella_pragathi/Downloads/gcc-15.1.0/bin:/mnt/c/Program Files/Git/cmd:/
mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/jella_pragathi/AppData/Local/Programs/Python/Python314/Scripts:/m
nt/c/Users/jella_pragathi/AppData/Local/Programs/Python/Python314:/mnt/c/Users/jella_pragathi/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/jella_pragat
hi/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/jella_pragathi/AppData/Roaming/npm:/snap/bin

Searching PATH...
Command not found: hello123
jella_pragathi@LAPTOP-RQQI:~$ ./build/task2_path /usr/bin/ls
=== PATH Command Resolution ===

Command to resolve: /usr/bin/ls

PATH variable:
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporat
ionII.WindowsSubsystemForLinux.2.7.11.0_x64_8wekyb3d8bbwe:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Or
acle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/windows/system32:/mnt/c/windows:/mnt/c/windows/System32/Wbem:/mnt/c/
windows/System32/WindowsPowerShell/v1.0:/mnt/c/windows/System32/OpenSSH:/mnt/c/Users/jella_pragathi/Downloads/gcc-15.1.0/bin:/mnt/c/Program Files/Git/cmd:/
mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Users/jella_pragathi/AppData/Local/Programs/Python/Python314/Scripts:/m
nt/c/Users/jella_pragathi/AppData/Local/Programs/Python/Python314:/mnt/c/Users/jella_pragathi/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/jella_pragat
hi/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/jella_pragathi/AppData/Roaming/npm:/snap/bin

Searching PATH...
Command found!
Executable path: /usr/bin/ls
jella_pragathi@LAPTOP-RQQI:~$
```

Session 8:

Task 1:

To Detect Variable References, Expand Values, Handle Undefined Variables, Update Tokens, Support Nested Variables, Test Expansion Lo



```
GNU nano 8.7.1 src/task1_variable.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>

#define MAX_INPUT 1024
#define MAX_OUTPUT 4096

void expand_variables(const char *input, char *output, size_t output_size)
{
    size_t i = 0;
    size_t j = 0;

    while (input[i] != '\0' && j < output_size - 1)
    {
        if (input[i] == '$')
        {
            char variable[256];
            size_t k = 0;
            const char *value;

            i++;

            /* Handle ${VARIABLE} */
            if (input[i] == '{')
            {
                i++;

                while (input[i] != '\0' &&
                    input[i] != '}' &&
                    k < sizeof(variable) - 1)
                {
                    variable[k++] = input[i++];
                }

                if (input[i] == '}')
                    i++;
            }
        }
        output[j++] = input[i++];
    }
}
```

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```
}
else
{
    /* Handle $VARIABLE */
    while (input[i] != '\0' &&
           (isalnum((unsigned char)input[i]) ||
            input[i] == '_') &&
           k < sizeof(variable) - 1)
    {
        variable[k++] = input[i++];
    }

    variable[k] = '\0';

    /* If there is no variable name after $ */
    if (k == 0)
    {
        output[j++] = '$';
        continue;
    }

    /* Get variable value from environment */
    value = getenv(variable);

    if (value != NULL)
    {
        while (*value != '\0' &&
               j < output_size - 1)
        {
            output[j++] = *value++;
        }
    }
    else
    {
        /* Undefined variables become empty */
    }
}
```

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```

        }
    }
    else
    {
        output[j++] = input[i++];
    }
}

output[j] = '\0';
}

int main(void)
{
    char input[MAX_INPUT];
    char output[MAX_OUTPUT];

    printf("=== Variable Expansion ===\n");
    printf("Enter a string: ");

    if (fgets(input, sizeof(input), stdin) == NULL)
        return 1;

    input[strcspn(input, "\n")] = '\0';

    expand_variables(input, output, sizeof(output));

    printf("Original : %s\n", input);
    printf("Expanded : %s\n", output);

    return 0;
}

```

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^{^U} Paste

^{^T} Execute
^{^J} Justify

```
jella_pragathi@LAPTOP-RQQG555:~/ossproject$ nano src/task1_variable.c
jella_pragathi@LAPTOP-RQQG555:~/ossproject$ gcc -Wall -Wextra -g src/task1_variable.c -o build/task1_variable
jella_pragathi@LAPTOP-RQQG555:~/ossproject$ ./build/task1_variable
=== Variable Expansion ===
Enter a string: Hello $USER
Original : Hello $USER
Expanded : Hello jella_pragathi
jella_pragathi@LAPTOP-RQQG555:~/ossproject$ ./build/task1_variable
=== Variable Expansion ===
Enter a string: Home directory: $HOME
Original : Home directory: $HOME
Expanded : Home directory: /home/jella_pragathi
jella_pragathi@LAPTOP-RQQG555:~/ossproject$ ./build/task1_variable
=== Variable Expansion ===
Enter a string: Path: ${HOME}/ossproject
Original : Path: ${HOME}/ossproject
Expanded : Path: /home/jella_pragathi/ossproject
jella_pragathi@LAPTOP-RQQG555:~/ossproject$ User=$USER Home=$HOME
jella_pragathi@LAPTOP-RQQG555:~/ossproject$ Value: $XYZ_NOT_DEFINED
Value:: command not found
jella_pragathi@LAPTOP-RQQG555:~/ossproject$ ./build/task1_variable
=== Variable Expansion ===
Enter a string: User=$USER Home=$HOME
Original : User=$USER Home=$HOME
Expanded : User=jella_pragathi Home=/home/jella_pragathi
jella_pragathi@LAPTOP-RQQG555:~/ossproject$ ./build/task1_variable
=== Variable Expansion ===
Enter a string: Value: $XYZ_NOT_DEFINED
Original : Value: $XYZ_NOT_DEFINED
Expanded : Value:
jella_pragathi@LAPTOP-RQQG555:~/ossproject$ export VAR1='$HOME'
export VAR2='$VAR1'
jella_pragathi@LAPTOP-RQQG555:~/ossproject$ echo $VAR1
$HOME
$VAR1
jella_pragathi@LAPTOP-RQQG555:~/ossproject$ ./build/task1_variable
=== Variable Expansion ===
Enter a string: Nested value: $VAR2
Original : Nested value: $VAR2
Expanded : Nested value: $VAR1
```

Task 2:

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic

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GNU nano 8.7.1

src/task2_dispatch.c *

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
```

```
#define MAX_INPUT 1024
#define MAX_ARGS 64
```

```
typedef void (*builtin_function)(char **args);
```

```
/* Built-in: pwd */
```

```
void builtin_pwd(char **args)
```

```
{
```

```
    char cwd[1024];
```

```
    (void)args;
```

```
    if (getcwd(cwd, sizeof(cwd)) != NULL)
```

```
    {
```

```
        printf("%s\n", cwd);
```

```
    }
```

```
    else
```

```
    {
```

```
        perror("pwd");
```

```
    }
```

```
}
```

```
/* Built-in: cd */
```

```
void builtin_cd(char **args)
```

```
{
```

```
    if (args[1] == NULL)
```

```
    {
```

```
        fprintf(stderr, "cd: missing argument\n");
```

```
        return;
```

```
    }
```

```
    if (chdir(args[1]) != 0)
```

```
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^T Execute
^J Justify
```

```
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GNU nano 8.7.1 src/task2_dispatch.c *
{
    perror("cd");
}
}

/* Built-in: echo */
void builtin_echo(char **args)
{
    int i = 1;

    while (args[i] != NULL)
    {
        printf("%s", args[i]);

        if (args[i + 1] != NULL)
        {
            printf(" ");
        }

        i++;
    }

    printf("\n");
}

/* Built-in: help */
void builtin_help(char **args)
{
    (void)args;

    printf("\nBuilt-in commands:\n");
    printf("  cd <directory>  Change current directory\n");
    printf("  pwd            Display current directory\n");
    printf("  echo <text>     Display text\n");
    printf("  help           Display this help\n");
    printf("  exit           Exit the shell\n");
}

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^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify
```

```
jella_pragathi@LAPTOP-RQQK: ~$ nano src/task2_dispatch.c
GNU nano 8.7.1 src/task2_dispatch.c *

/* Built-in: exit */
void builtin_exit(char **args)
{
    (void)args;

    printf("Exiting shell...\n");
    exit(0);
}

/*
 * Dispatch table
 */
typedef struct
{
    const char *name;
    builtin_function function;
} Builtin;

Builtin builtin_table[] =
{
    {"cd", builtin_cd},
    {"pwd", builtin_pwd},
    {"echo", builtin_echo},
    {"help", builtin_help},
    {"exit", builtin_exit}
};

#define BUILTIN_COUNT \
    (sizeof(builtin_table) / sizeof(builtin_table[0]))

/*
 * Identify and execute a built-in command.
 */
int dispatch_builtin(char **args)
{
    if (args[0] == NULL)

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```

```
jella_pragathi@LAPTOP-RQQ( × + ▾
GNU nano 8.7.1 src/task2_dispatch.c *
int dispatch_builtin(char **args)
{
    if (args[0] == NULL)
    {
        return 1;
    }

    for (size_t i = 0; i < BUILTIN_COUNT; i++)
    {
        if (strcmp(args[0], builtin_table[i].name) == 0)
        {
            builtin_table[i].function(args);
            return 1;
        }
    }

    return 0;
}

/*
 * Split input into tokens.
 */
int parse_command(char *input, char **args)
{
    int count = 0;

    char *token = strtok(input, " \\t");

    while (token != NULL && count < MAX_ARGS - 1)
    {
        args[count++] = token;
        token = strtok(NULL, " \\t");
    }

    args[count] = NULL;

    return count;
}

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```

```
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GNU nano 8.7.1 src/task2_dispatch.c *
    return count;
}

int main(void)
{
    char input[MAX_INPUT];
    char *args[MAX_ARGS];

    printf("=== Built-in Command Dispatch ===\n");
    printf("Type 'help' to see available commands.\n");

    while (1)
    {
        printf("dispatch-shell> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL)
        {
            break;
        }

        input[strcspn(input, "\n")] = '\0';

        if (parse_command(input, args) == 0)
        {
            continue;
        }

        if (!dispatch_builtin(args))
        {
            printf("Invalid built-in command: %s\n", args[0]);
        }
    }

    return 0;
}
|
```

Help	Write Out	Where Is	Cut	Execute
Exit	Read File	Replace	Paste	Justify

```
jella_pragathi@LAPTOP-RQQGT555:~$ cd ~/ossp_project
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ pwd
/home/jella_pragathi/ossp_project
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ nano src/task2_dispatch.c
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ gcc -Wall -Wextra -g src/task2_dispatch.c -o build/task2_dispatch
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ ./build/task2_dispatch
=== Built-in Command Dispatch ===
Type 'help' to see available commands.
dispatch-shell> help

Built-in commands:
  cd <directory>  Change current directory
  pwd             Display current directory
  echo <text>     Display text
  help           Display this help
  exit           Exit the shell

dispatch-shell> pwd
/home/jella_pragathi/ossp_project
dispatch-shell> echo Hello World
Hello World
dispatch-shell> pwd
/home/jella_pragathi/ossp_project
dispatch-shell> cd /tmp
dispatch-shell> pwd
/tmp
dispatch-shell> hello
Invalid built-in command: hello
dispatch-shell> cd
cd: missing argument
dispatch-shell> exit
Exiting shell...
jella_pragathi@LAPTOP-RQQGT555:~/ossp_project$ |
```